Deep Learning Optimized on Jean Zay

Dataset optimization

Storage spaces and data format

cnrs IDRIS

Dataset optimization

Main bottlenecks ◂

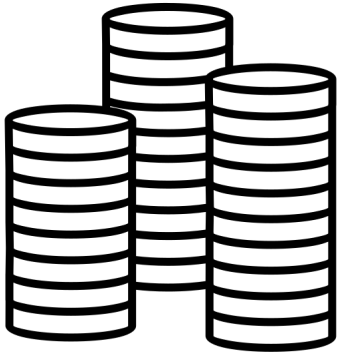
Data storage – various disk spaces ◂

Data format – at sample level ◂

Data format – at dataset level ◂

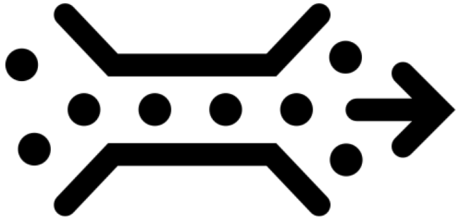
Bottlenecks upstream of DataLoader

Storage disks



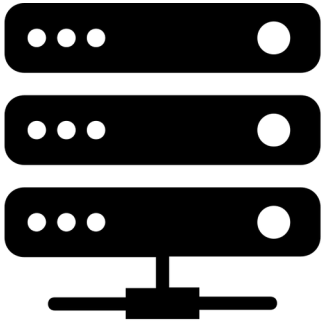
1. I/O performance

Interconnection network
Omnipath/InfiniBand



2. Shared bandwidth

CPU workers



3. Decoder performance

Bottlenecks upstream of DataLoader

Storage disks

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Interconnection network
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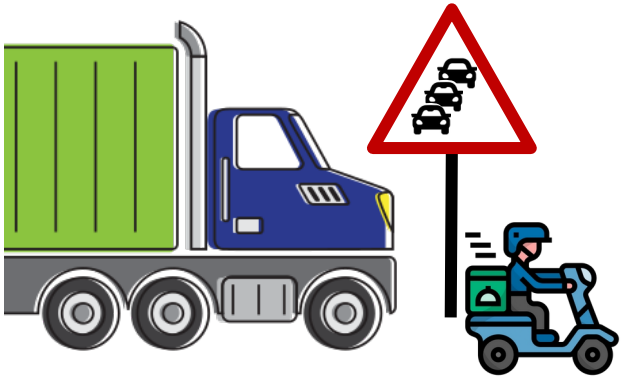

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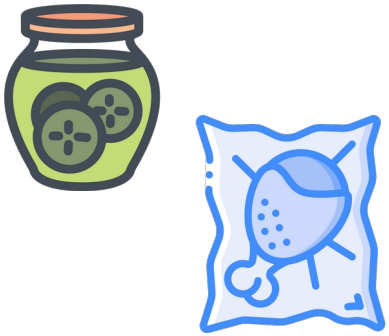

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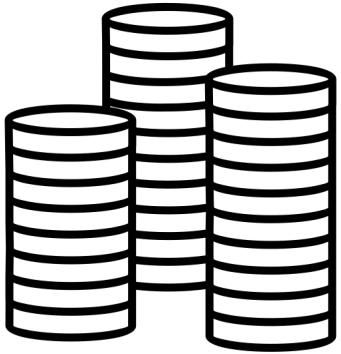

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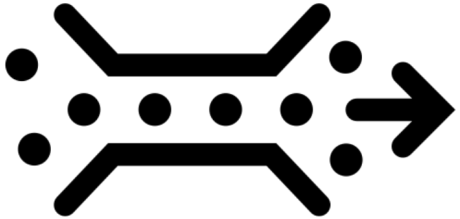

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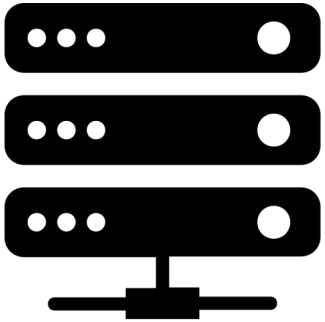
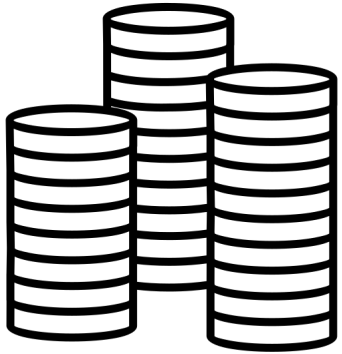


3. Decoder performance

Where should I store my dataset?

Various disk spaces

Storage disks



1. I/O performance

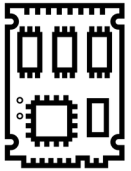
WORK

Rotative disk spaces

300-350 GB/s

SCRATCH

Full Flash



1.1-1.5 TB/s

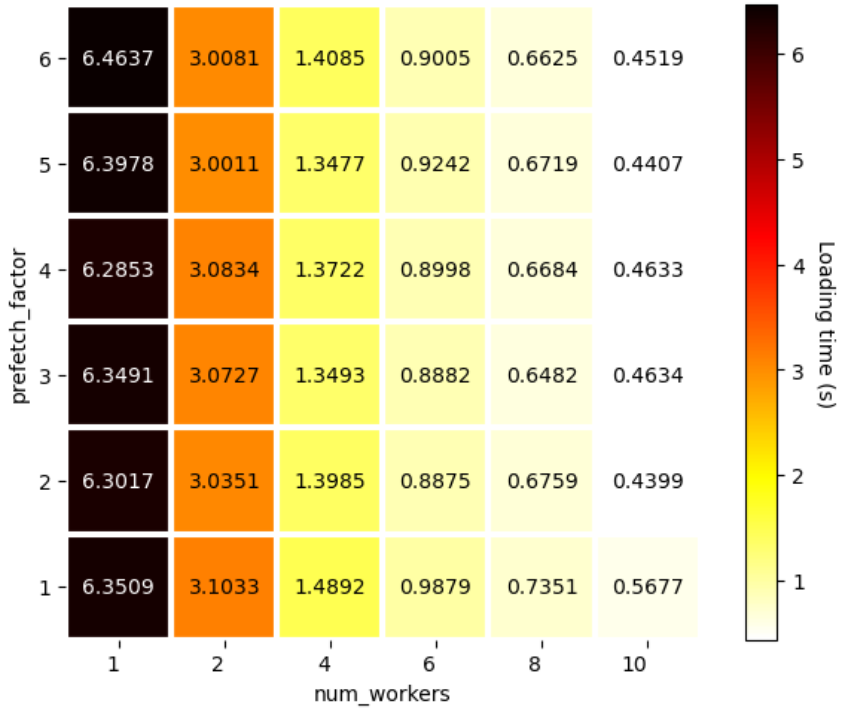
NVMe

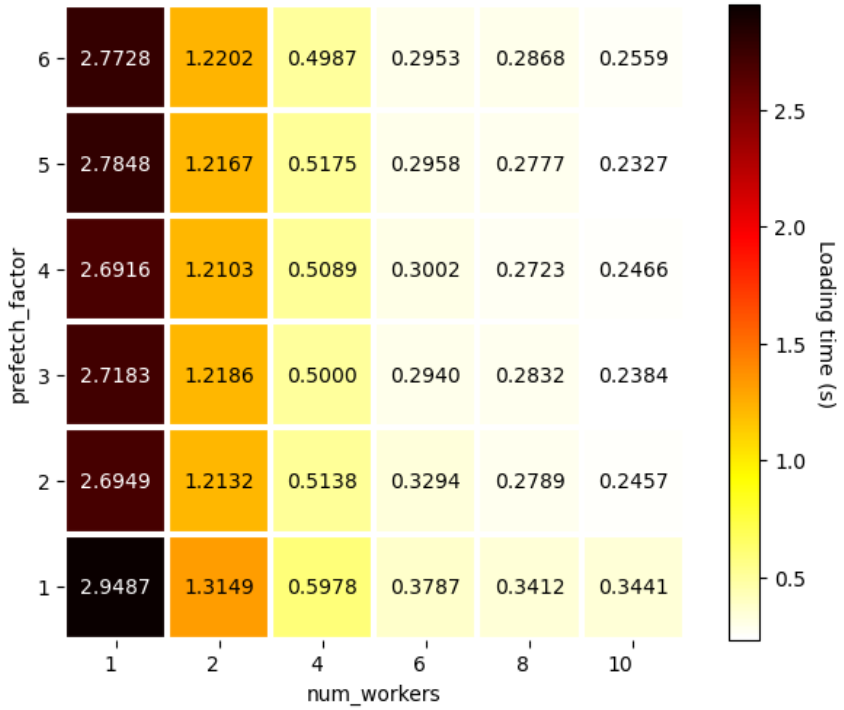
Local disk
(test configuration)

Local

Various disk spaces

dlojz.py - 50 iterations - test partition gpu_p4





WORK

100 GB/s - OPA

SCRATCH

500 GB/s - OPA

Various disk spaces

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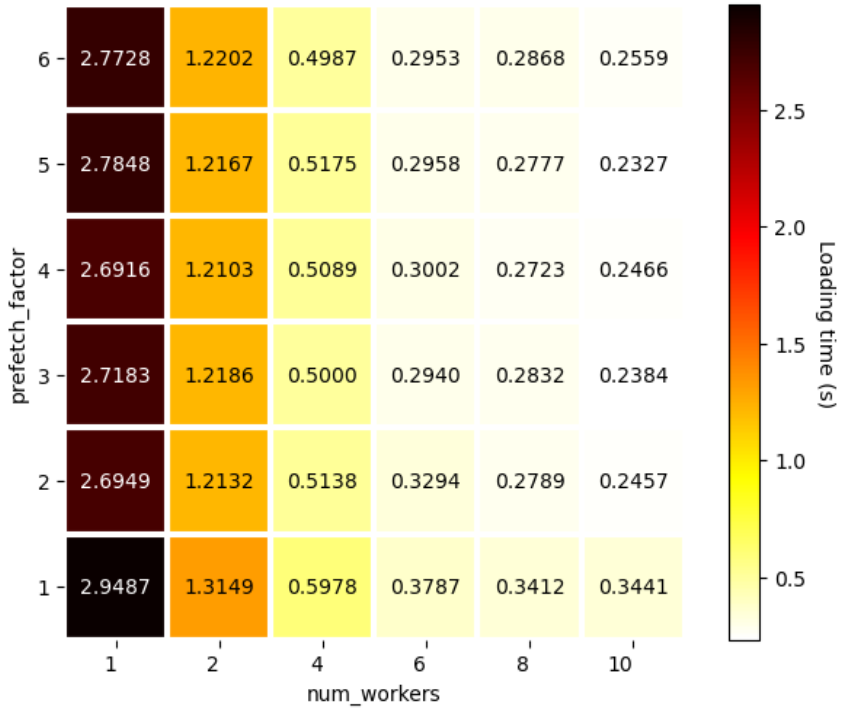

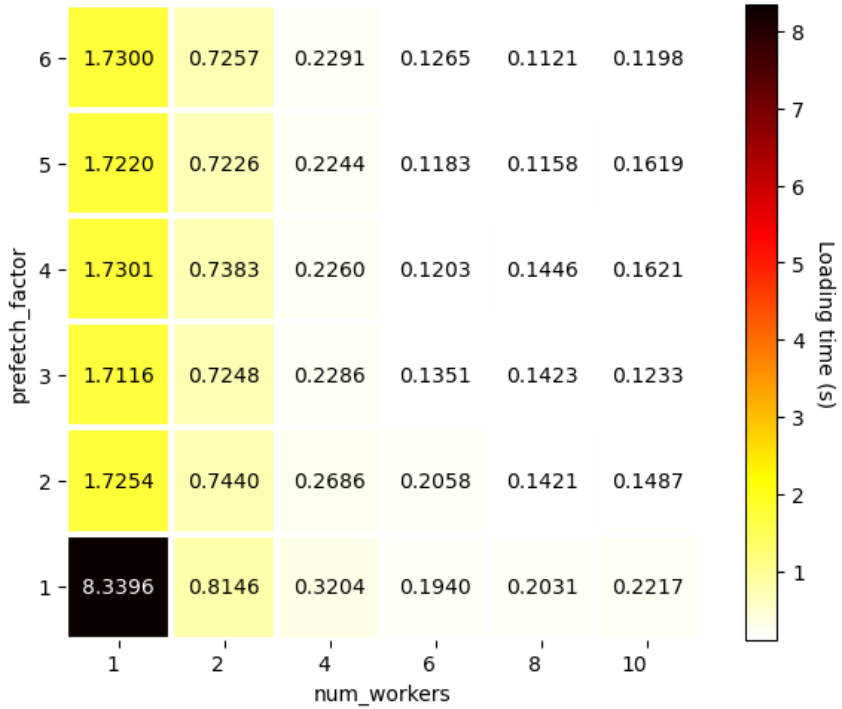

SCRATCH

500 GB/s - OPA

NVMe

PCIe

Various disk spaces

- **NVMe**
  - ✓ Best IO performance
  - – You need to copy your dataset on the local disk first, which can take a very long time
  - – This solution is not suitable at the scale of a supercomputer so it is not available to users

- **SCRATCH**
  - ✓ Second best IO performance
  - ✓ Very large quota (bytes and inodes)
  - – 30 days file lifespan

- **WORK**
  - – Worst performance (but not bad)
  - – Only 5 TB and 500k inodes
  - ✓ No automatic deletion procedure

Various disk spaces

- **What about the STORE?**
  - Magnetic tape storage (with a cache on rotative disk)
  - 50 TB and 100k inodes
  - Intended to receive few very large-sized files (up to 10 TiB per file) → **archives**
  - Not accessible from the compute nodes (very high read/write latency)

- **Good practice**
  - If you don’t need IO performance (when building your workflow, debugging, or simply if IO is not a bottleneck), store your dataset in the **WORK** if it is not too large
  - If you need IO performance or if it is too large, store your dataset in the **SCRATCH**
  - Keep a duplicate of your dataset **as an archive in the STORE**

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Bottlenecks upstream of DataLoader

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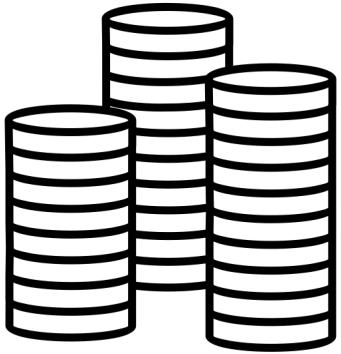

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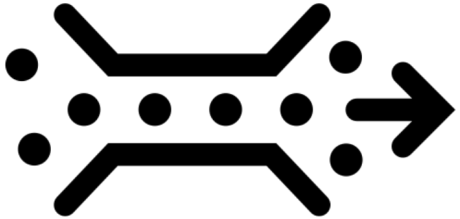
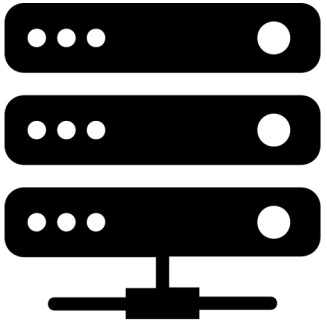


2. Shared bandwidth

CPU workers



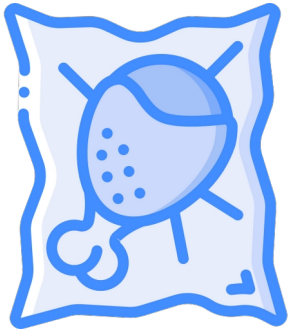
3. Decoder performance

Which format for my data?

At sample level - Sample decoding

Binary format: Pickle format, HDF5,...
Decoded more quickly, takes more space

- ✓ Decoder performance
- – Shared bandwidth
- – Storage volume



Compressed format: jpeg, png,...
Decoded more slowly, takes less space

- – Decoder performance
- ✓ Shared bandwidth
- ✓ Storage volume

Dataset optimisation

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Data format – at dataset level ◂

Bottlenecks upstream of DataLoader

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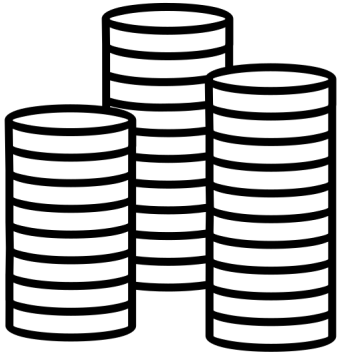

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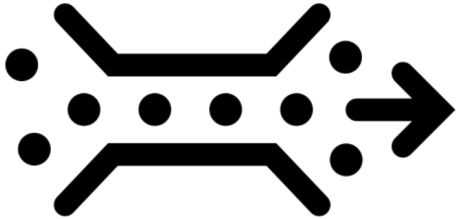

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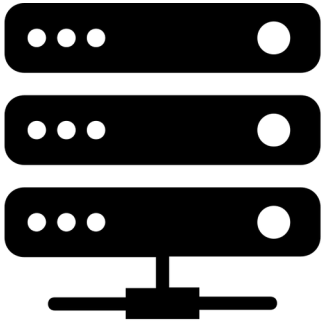
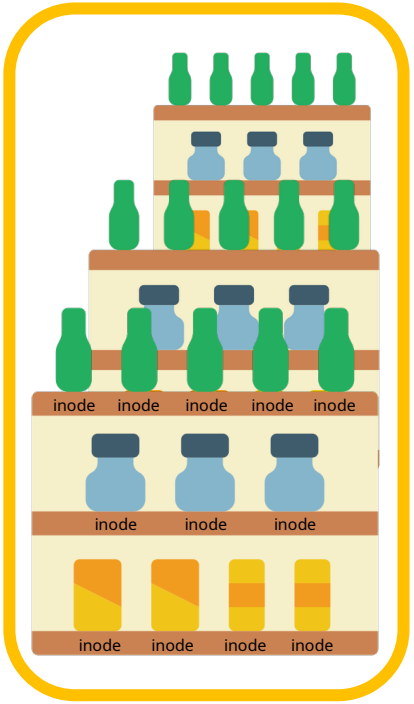


3. Decoder performance

Which format for my dataset?

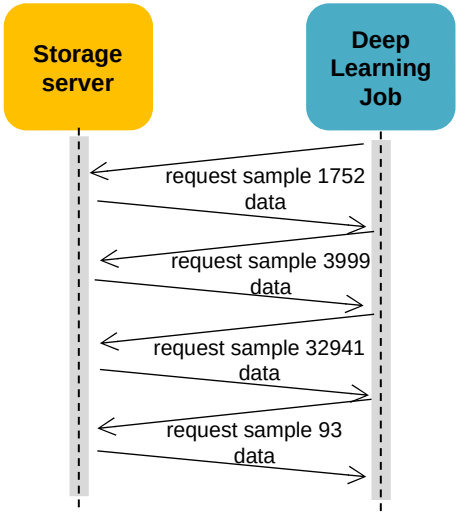
Intuitive way: 1 sample = 1 file

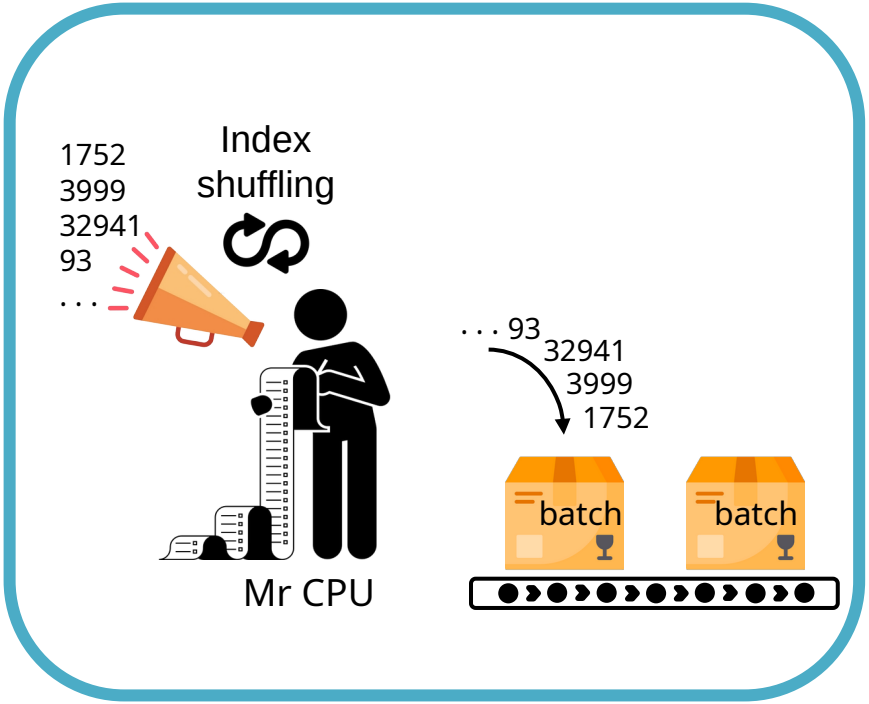


Map-style dataset

__getitem__

Random Access to File Store





Pros: Easy to handle, random access possible

Cons: Lots of inodes, lots of I/Os

Too many inodes is an issue

Error: Disk quota exceeded

- $WORK quota per user is 5 TB / **500 kinodes**
- $SCRATCH safety quota per user is ~500TB / **150 Minodes**

+ Parallel file system does not like:

- intensive IO workloads involving lots of small files
- large number of files in a single directory

Too many inodes is an issue

- IDRIS recommendation on Jean Zay: **less than 100k inodes per folder**

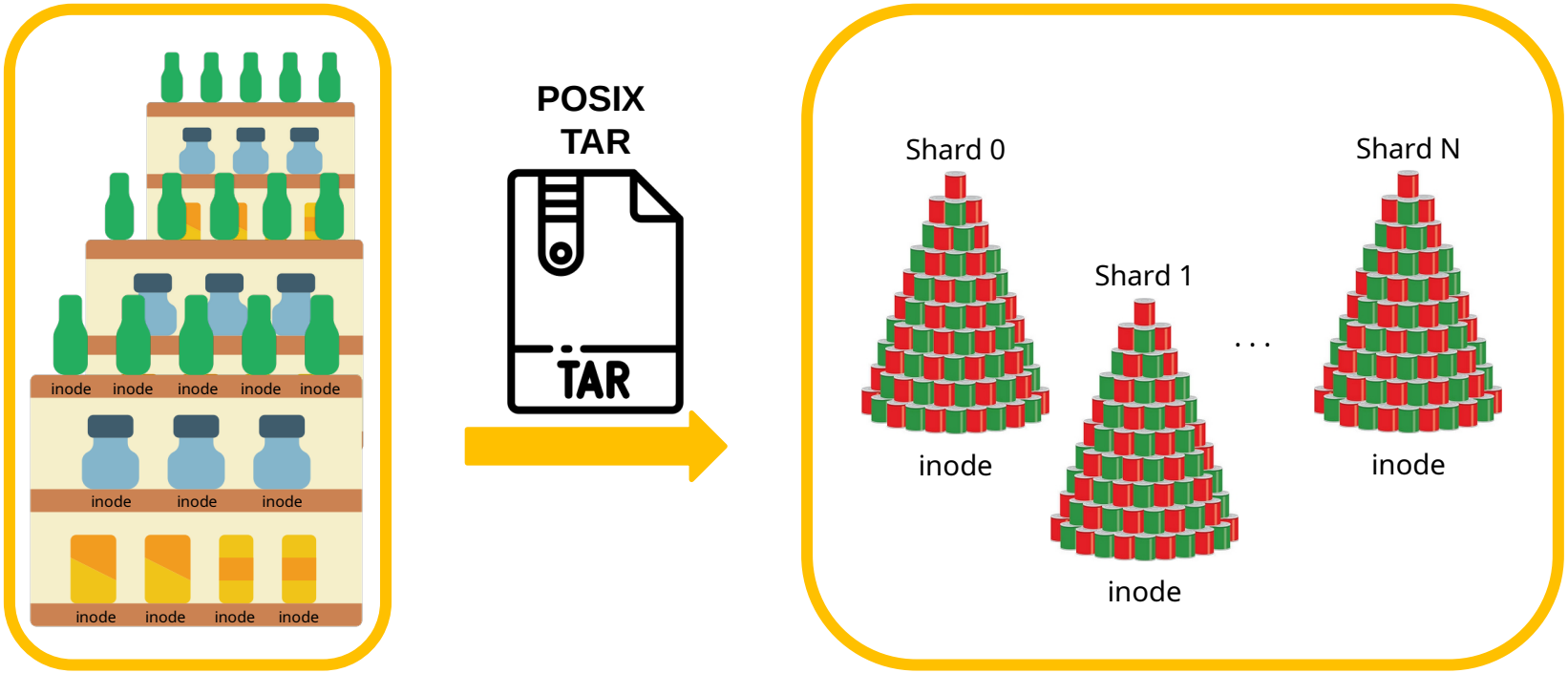
- Repository limitations and recommendations on HuggingFace Hub

Characteristics	Recommended	Tips
Files per repo	<100k	merge data into fewer files
Entries per folder	<10k	use subdirectories in repo

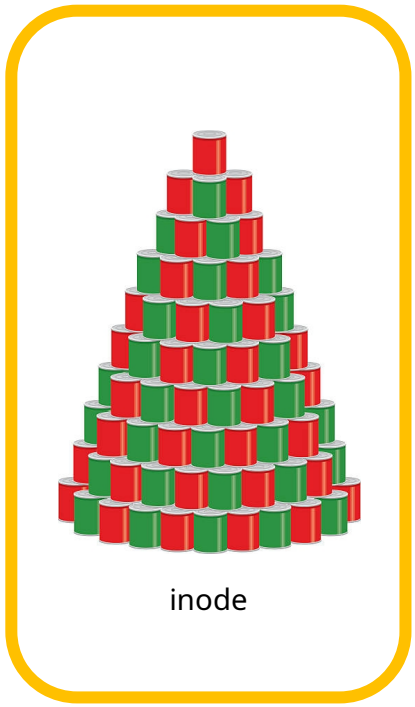
https://huggingface.co/docs/hub/storage-limits#repository-limitations-and-recommendations

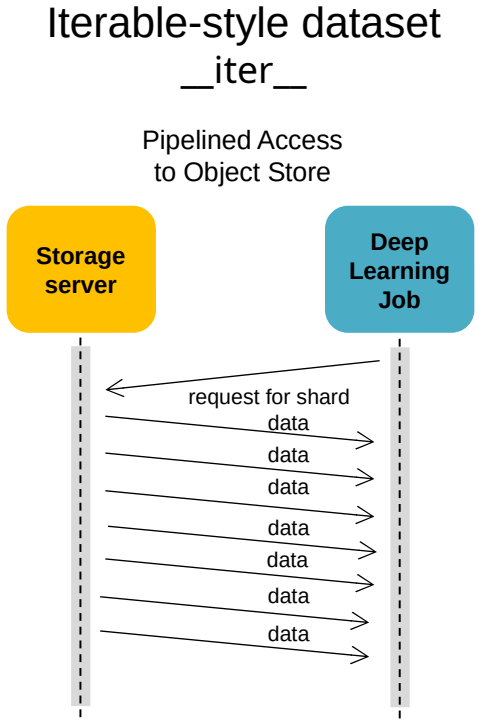
+ for large datasets, **WebDataset** format is recommended

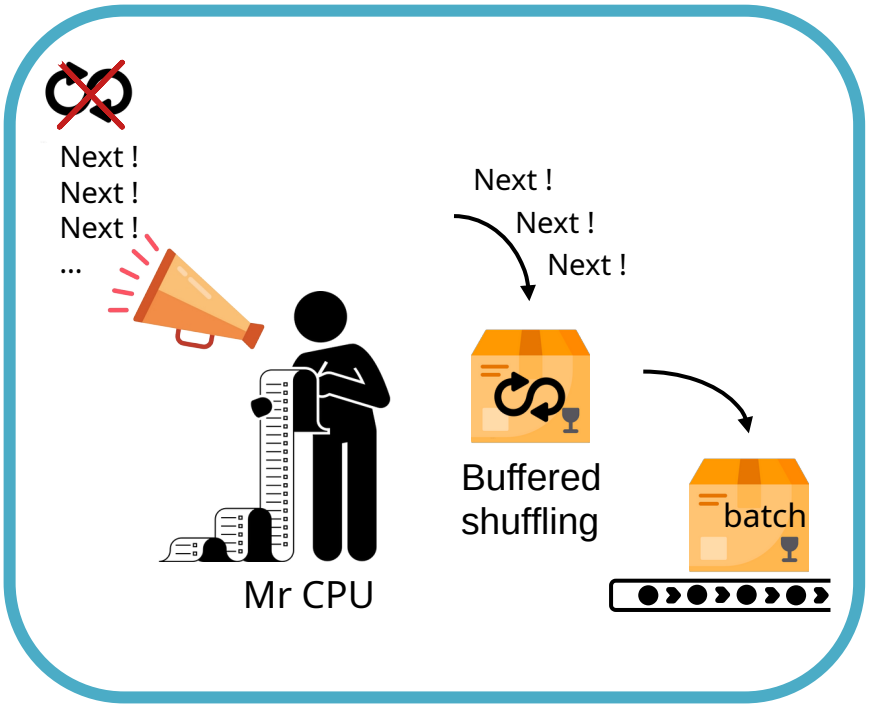
WebDataset format – Gathering data



WebDataset format – Iterable dataset





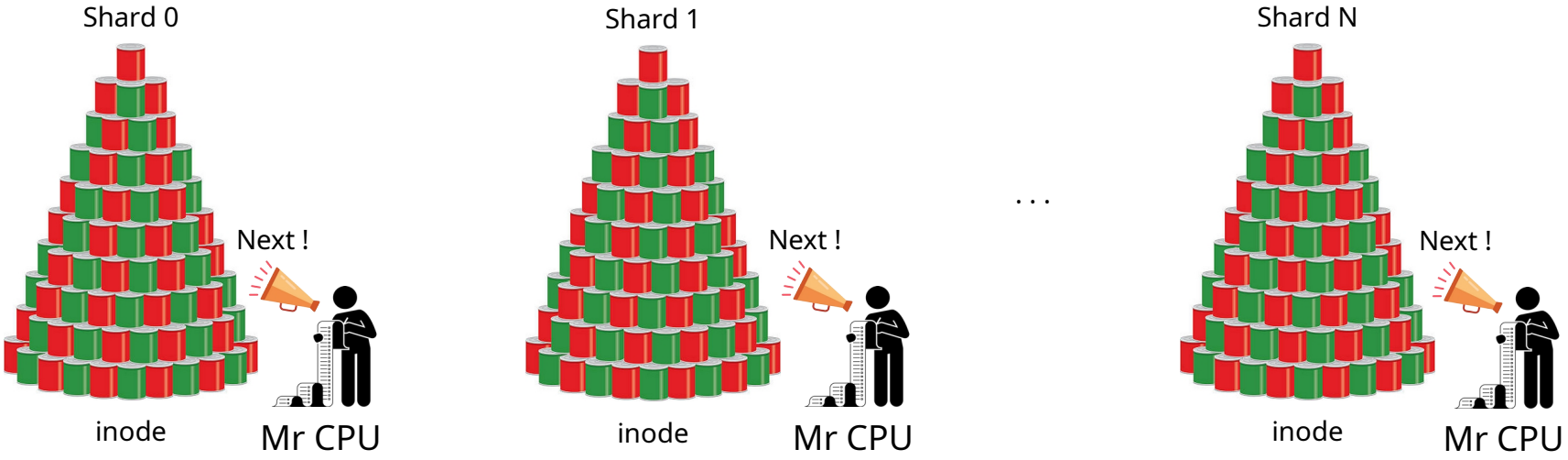


Pros: Fewer I/Os, fewer inodes

Cons: **Preprocessing**, difficult to shuffle or distribute, unknown dataset length

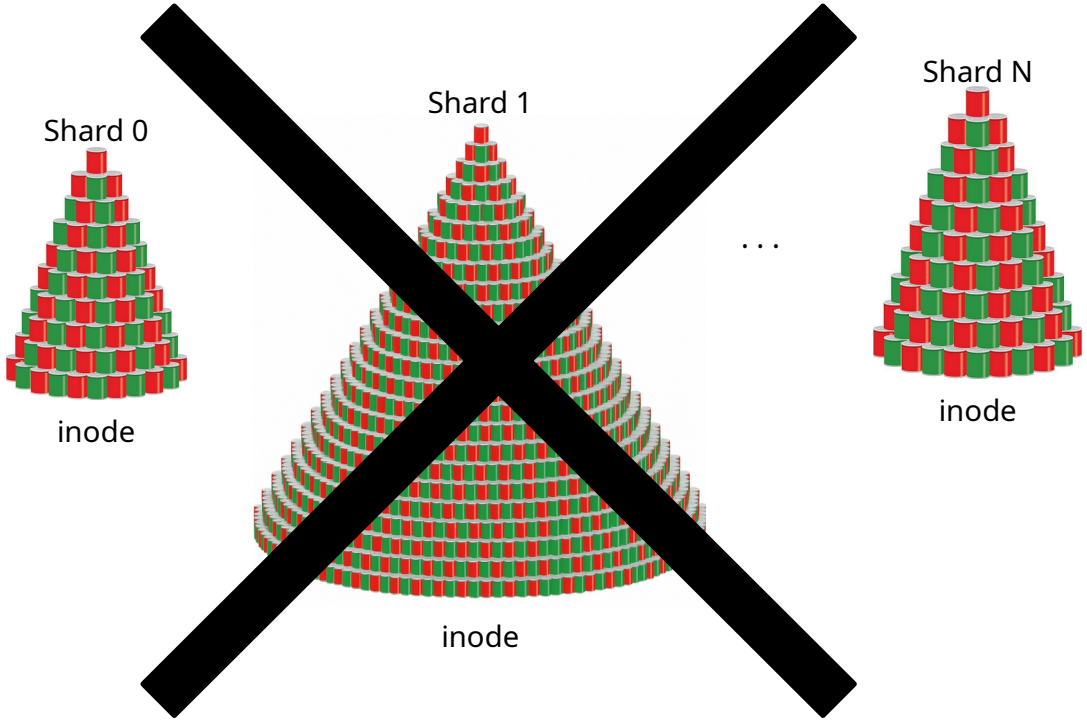
WebDataset format - Sharding

Sharding is necessary to benefit from parallel implementation
(DataLoader multi-processing and Distributed Data Parallelism).



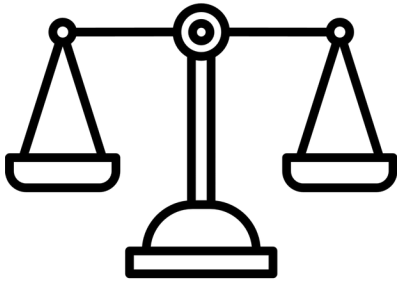
The number of shards should be a multiple of the number of tasks/GPUs.

WebDataset format - Sharding

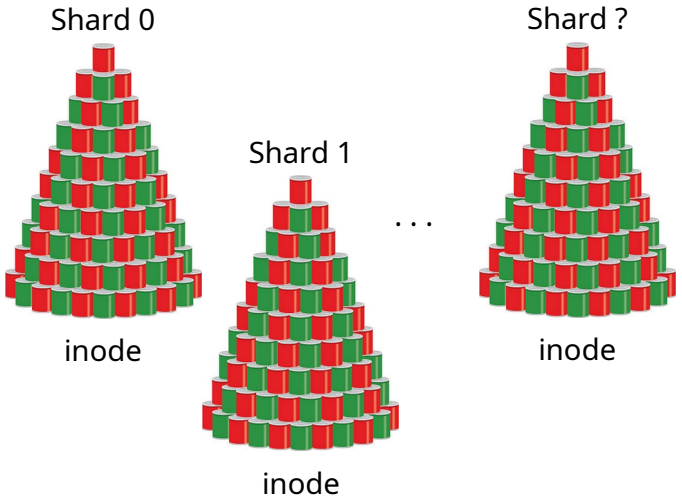


Samples must be evenly distributed among the shards to balance the workload between processes.

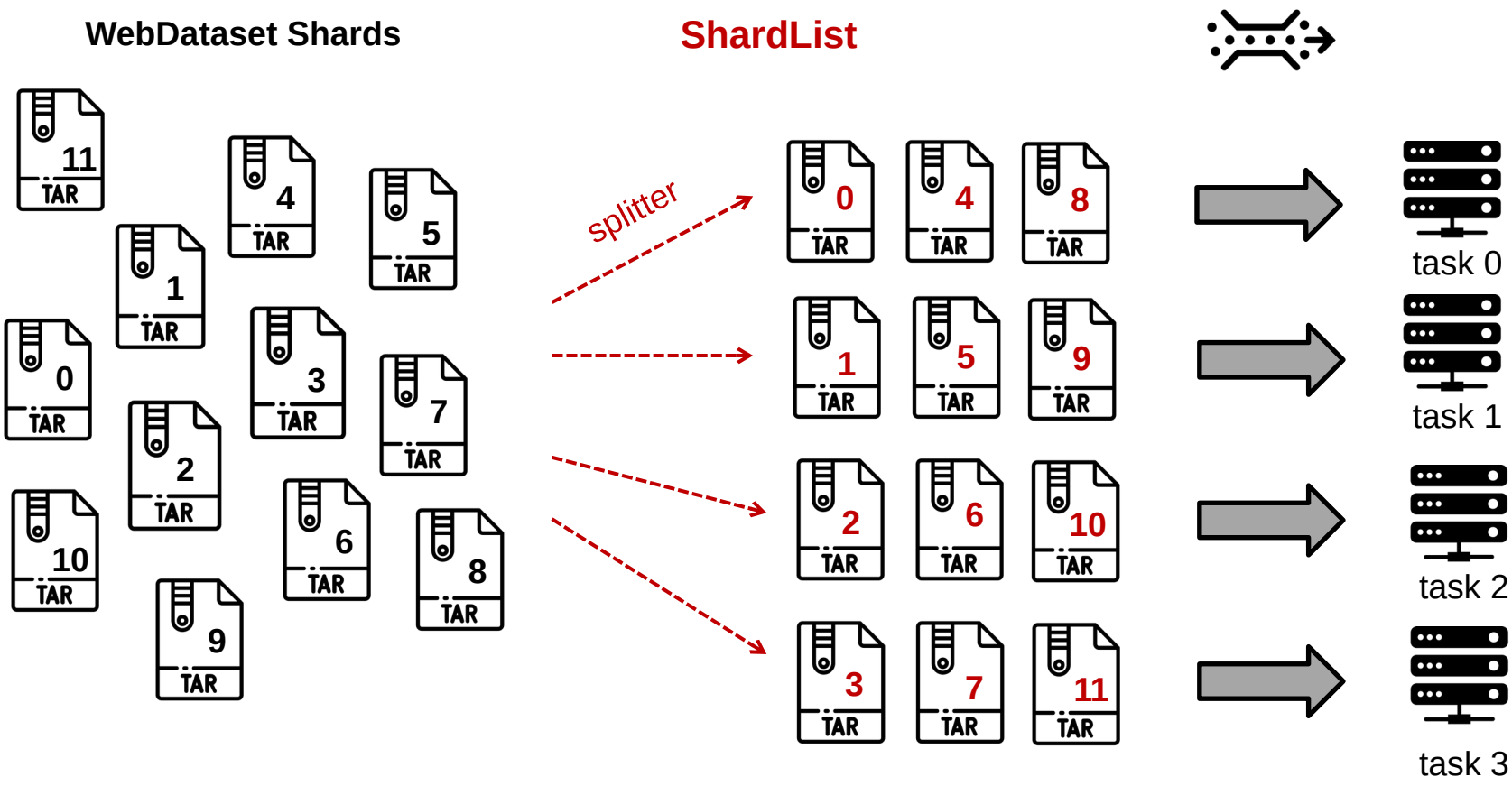
WebDataset format - More or less shards?



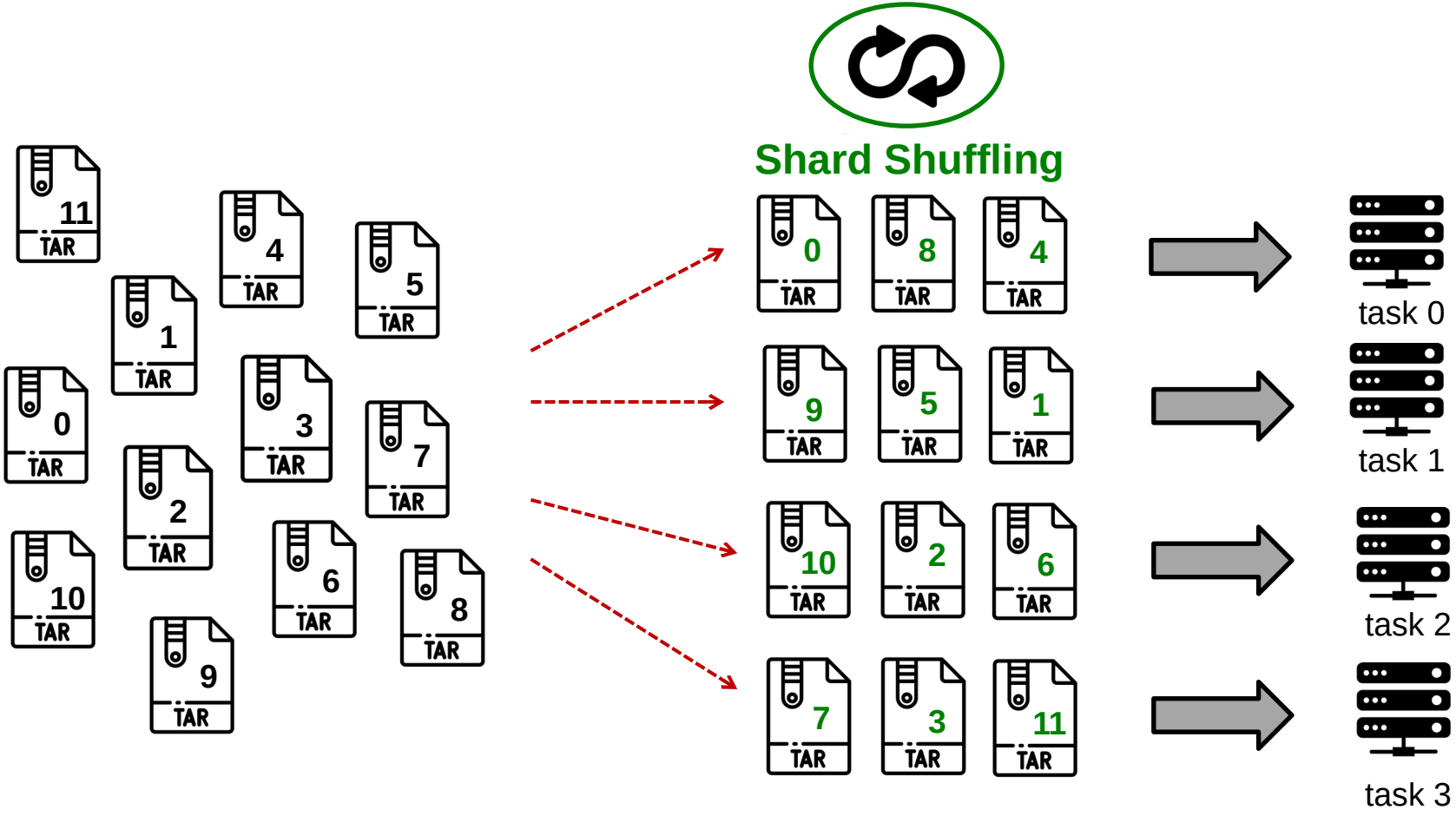
	More shards	Less shards
Large scale distribution	+	-
Shared bandwidth	+	-
Inodes quota	-	+
Number of I/O	-	+



WebDataset – Multiworker sharding



WebDataset - Shuffling



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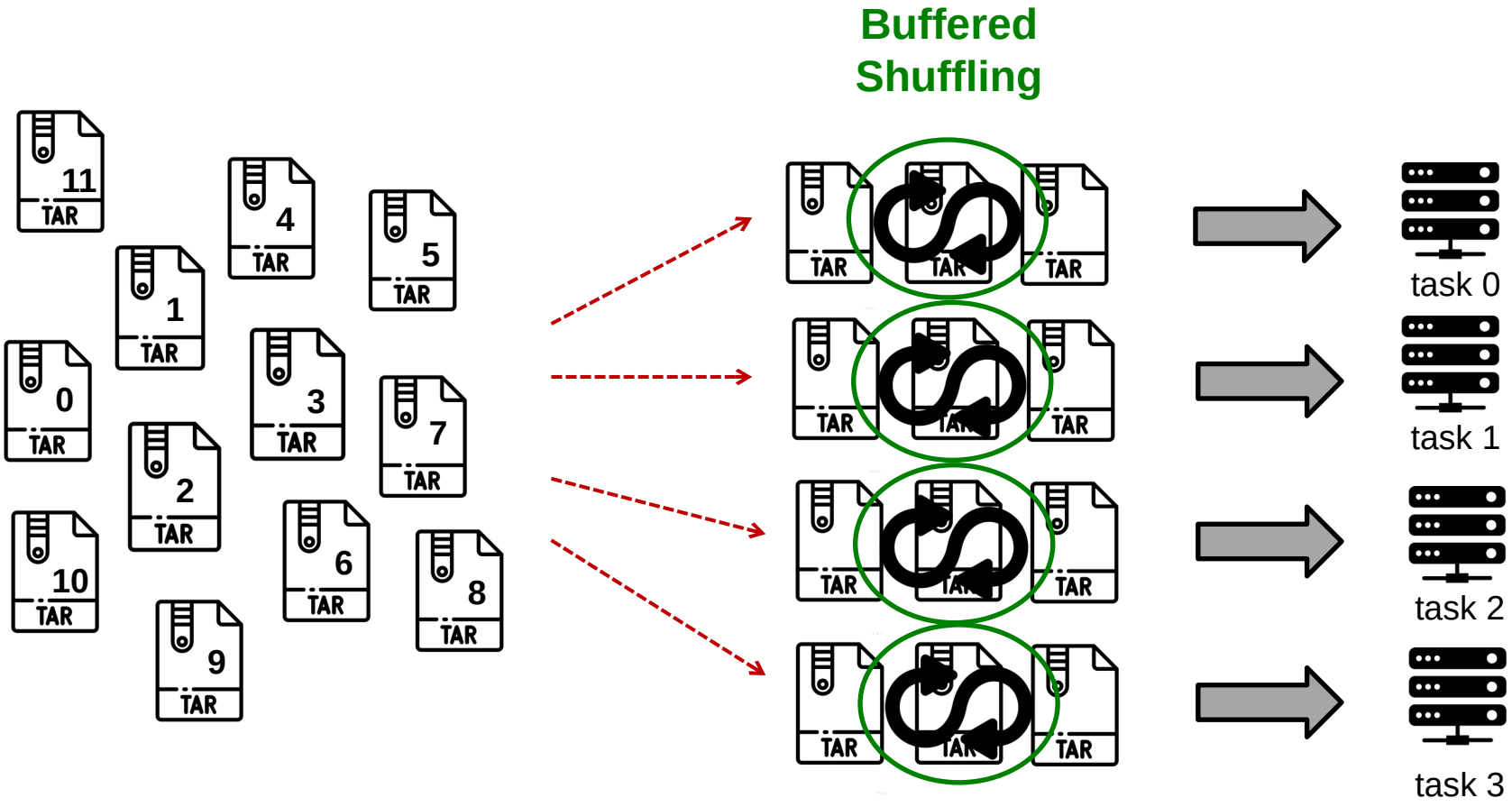
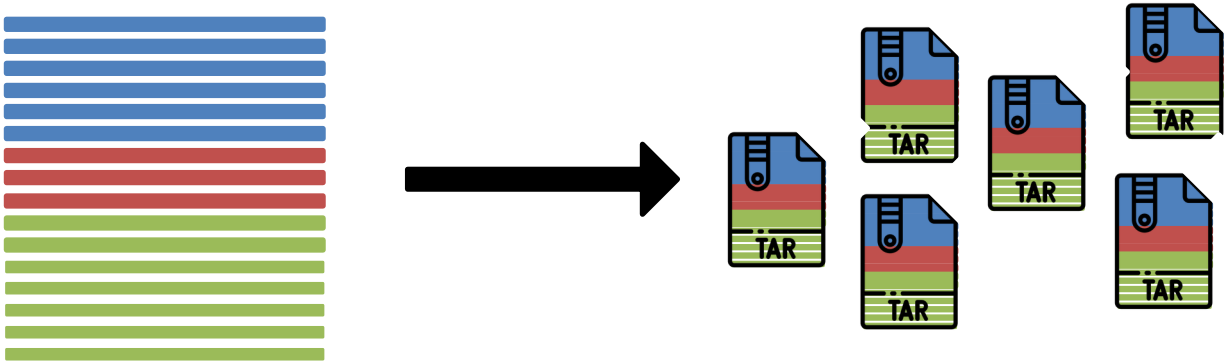


WebDataset - Preprocessing

- Distribute the samples as **evenly** as possible among the shards.
- Choose the number of shards **according to the number of GPUs & DataLoader workers** you will use.
- Distribute the samples so that each shard contains a **representative part of the dataset.**



+ Converting data before creating the archives to improve decoding performance?

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
                                .shuffle(1000)\
                                .decode("torchrgb")\
                                .to_tuple('input.pyd', 'output.pyd')\
                                .map_tuple(transform, lambda x: x)\
                                .batched(mini_batch_size)\
                                .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
                             num_workers=num_workers,\
                             persistent_workers=persistent_workers,\
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                             ).slice(nbatches)
train_loader.length = nbatches
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distribute shards among processes

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shuffling shards indexes per process

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                                .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
                             num_workers=num_workers,\
                             persistent_workers=persistent_workers,\
                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

shuffling samples per process

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
                                .shuffle(1000)\
                                .decode("torchrgb")\
                                .to_tuple('input.pyd', 'output.pyd')\
                                .map_tuple(transform, lambda x: x)\
                                .batched(mini_batch_size)\
                                .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
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                             num_workers=num_workers,\
                             persistent_workers=persistent_workers,\
                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

} description of shard content

WebDataset - Implementation

```
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                             num_workers=num_workers,\
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                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

transforming and batching

WebDataset - Implementation

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                             num_workers=num_workers,\
                             persistent_workers=persistent_workers,\
                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

define len(train_dataset)

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
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                             num_workers=num_workers,\
                             persistent_workers=persistent_workers,\
                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

batching handled by WebDataset class

WebDataset - Implementation

```
import webdataset as wds

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    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

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                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

usual DataLoader args

WebDataset - Implementation

```
import webdataset as wds

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    return paths[idr_torch.rank::idr_torch.world_size]

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                             persistent_workers=persistent_workers,\
                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

drop_last equivalent

WebDataset - Implementation

```
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                                .to_tuple('input.pyd', 'output.pyd')\
                                .map_tuple(transform, lambda x: x)\
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                             num_workers=num_workers,\
                             persistent_workers=persistent_workers,\
                             pin_memory=pin_memory,\
                             prefetch_factor=prefetch_factor\
                             ).slice(nbatches)
train_loader.length = nbatches
```

define len(train_loader)

WebDataset - Perfomance test

I/O loop over the dataset
(calculation-free iterations)

```
start_time = datetime.datetime.now()

for i, (images,labels) in enumerate(loader):
    print(f'{i} / {nb_batches}', end="\r")

end_time = datetime.datetime.now()
delta_time = (end_time_it - start_time_it).total_seconds()
```

- Execution on 1 GPU

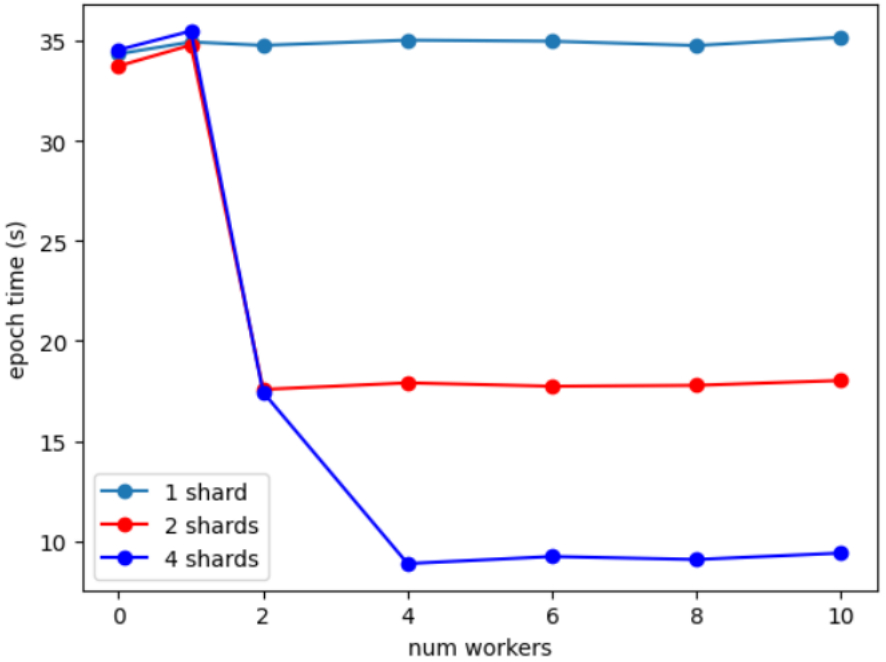
WebDataset - Perfomance test

I/O loop over the dataset
(calculation-free iterations)

CIFAR10 ~ 50k images

- Sharding is necessary to benefit from parallel implementation (DataLoader multi-processing).

CIFAR 10 (50k images)
elapsed time - 1 epoch

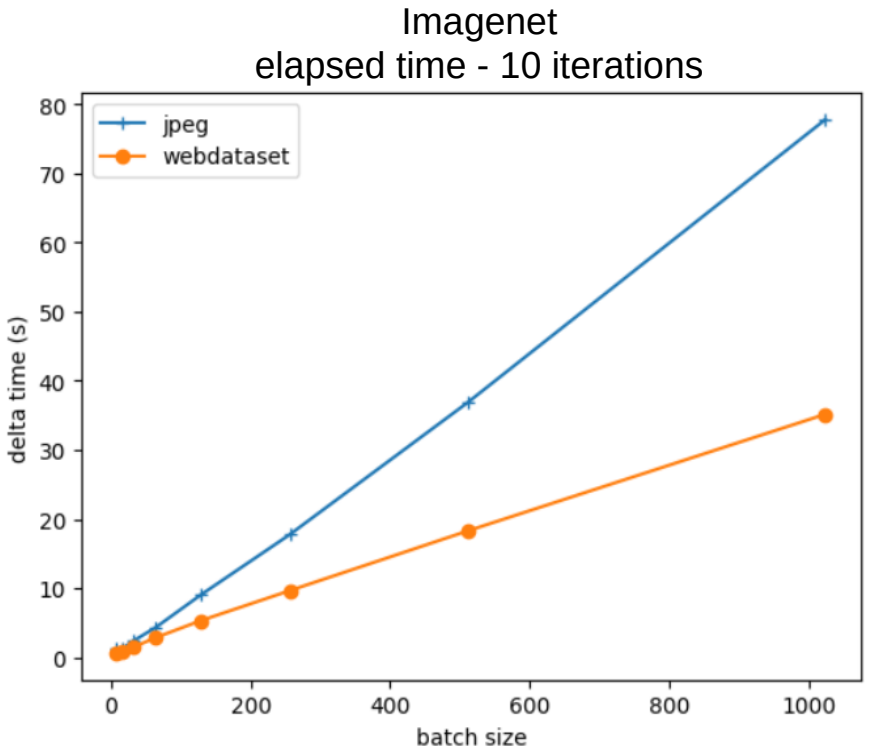


WebDataset - Perfomance test

I/O loop over the dataset
(calculation-free iterations)

Imagenet ~ 1.3M images
128 shards ~10k images per shard (+labels)
1 shard (images + labels) ~ 6GB

- The more samples are needed per batch, the more efficient is the WebDataset format (fewer I/Os).



more I/O generated

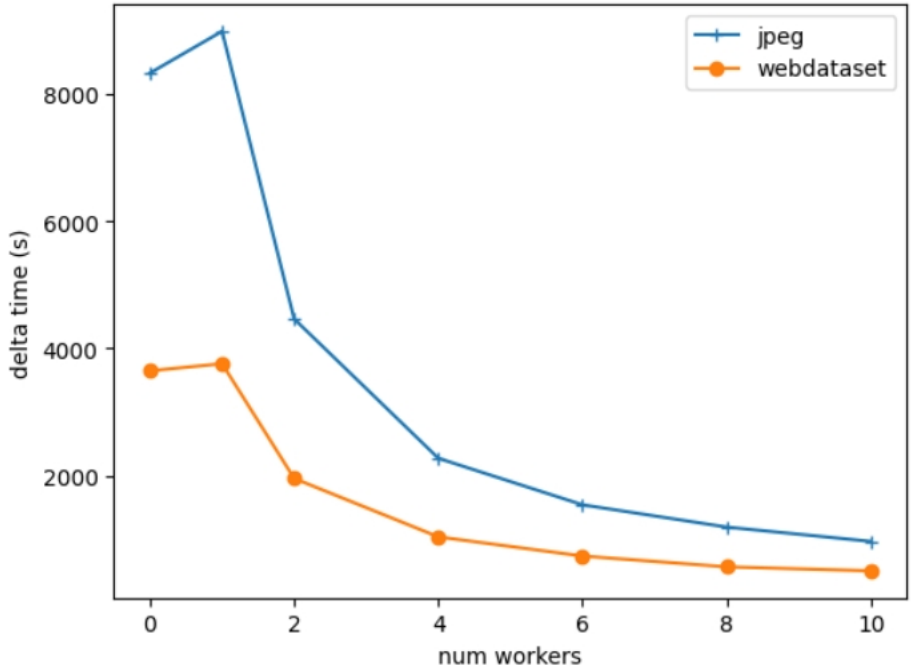
WebDataset - Perfomance test

I/O loop over the dataset
(calculation-free iterations)

Imagenet ~ 1.3M images
128 shards ~10k images per shard (+labels)
1 shard (images + labels) ~ 2GB

- The WebDataset format scales up.

Imagenet
elapsed time - 1 epoch

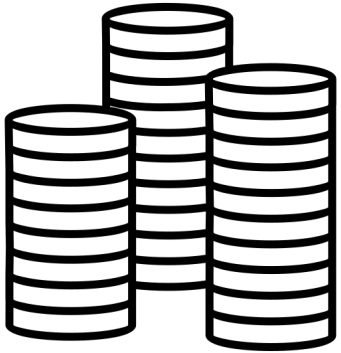


Conclusion on dataset format

- The more inodes your dataset contains, the more IOs you will request during training, and the more difficulty the parallel file system will have
- For large datasets, gather several samples in archives to minimize the number of inodes
- Standard formats already exist like WebDataset (tar files)

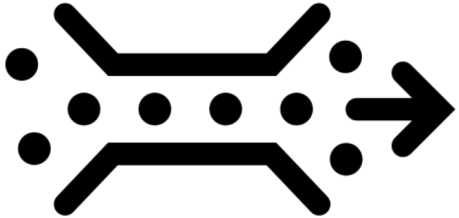
Conclusion

Storage disks



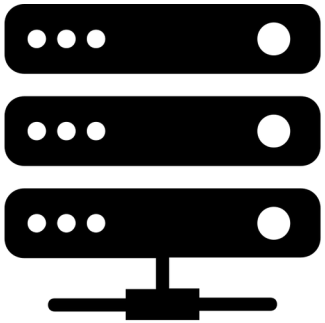
1. I/O performance

Interconnection network
Omnipath/InfiniBand



2. Shared bandwidth

CPU workers



3. Decoder performance

- Disk spaces: WORK or SCRATCH ?
- Data format: binary or compressed ?
- Dataset format: WebDataset format ?